

OpenClaw OLED Display Tutorial

Before starting, please complete the [01-Hardware Configuration], [02-openclaw API-KEY Configuration], [03-openclaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solution Configuration Tests] under the `Prerequisites` directory. If you are currently only using TUI / whatsapp, you can skip `04` for now. This article assumes the basic environment has been prepared and only expands on OLED-related content.

1. Tutorial Objectives

This article demonstrates OpenClaw's natural language control capabilities through OLED peripherals. After completing this tutorial, readers can:

- Understand the basic approach to controlling OLED display with OpenClaw
- Communicate with OpenClaw through three methods: whatsapp, TUI, and voice code
- Complete four OLED cases: text display, indicator display, weather card, animation display
- Understand the complete chain from "user input" to "screen display"

2. Hardware Preparation

- OpenClaw mainboard
- OLED display `[Default SSD1306 128x32 / I2C address 0x3C]`
- `[Optional] Microphone module`
- Wiring diagram



3. Software Preparation

Code entry points that will be used in the tutorial

- Low-level driver code: `/home/jetson/hardware_hal_demo/`
- OLED control command: `/home/jetson/hardware_hal_demo/oled`
- Temperature and humidity linked display: `/home/jetson/hardware_hal_demo/dht_oled`
- Voice entry: `/home/jetson/voice_to_openc1aw/src/main.py`

The `oled` tool already supports the following capabilities:

- Display text `text`
- Display image `image`

- Display animation `animation`
- Display weather card `weather`
- Clear screen `clear`

If you just want to quickly display temperature and humidity readings on OLED, you can also directly use `dht_oled`.

```
dht_oled #Display temperature and humidity data to OLED
oled text --text 'Hello, World' #Display text
oled weather --location 'Shenzhen' --condition 'Sunny' --temperature '26°C' --humidity '57%'
#Weather card
oled image --path ~/hardware_hal_demo/assets/oled/mountain_photo_128x32.png #Display image
oled animation --path ~/hardware_hal_demo/assets/oled/sunrise_anim_128x32.gif #Display GIF
animation
oled clear # Clear screen
```

Note: For GIF and image display, you can ask AI to generate 128x32 resolution images, then place them in this path: `~/hardware_hal_demo/assets/oled/`

More commands can be found in the md document in this folder,

```
/home/jetson/hardware_hal_demo/README.md
```

If underlying commands can execute normally, it indicates hardware and Jetson mainboard connection is normal and communication is normal.

4. Test OLED Display

Before starting this tutorial, please complete one `openclaw tui` basic dialogue self-check; if you haven't done it yet, see [05-Three Dialogue Solution Configuration Tests]. After confirming OpenClaw can respond normally, enter OLED-specific testing.

Terminal input to enter TUI:

```
openclaw tui
```

```

[11] OpenClaw 2026.3.12 (6472949)
Your terminal just grew claws—type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)

```

You can try inputting the following content in the terminal:

- Display "Hello, OpenClaw" on OLED
- Display "Today we start learning OLED " on screen
- Clear OLED screen
- Display current status "System OK" on OLED

Display "Hi, OpenClaw" on OLED

"Hi, OpenClaw" is now displayed on the OLED screen. ✓

Display "Today we start learning OLED" on the screen

"Today we start learning OLED" is now displayed on the screen. The text was automatically wrapped into two lines to fit the OLED display. ✓

Clear the OLED screen

OLED screen cleared. ✓

```

connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
46k/1.0m (5%)

```

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
whatsapp	Remote control, classroom demonstrations	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First run through the control chain
Voice code	Voice interaction, complete AI experience	Closest to real conversation experience	As final demonstration solution

If optional AI voice module + speaker are selected, whatsapp dialogue, tui, whatsapp dialogue need to broadcast OpenClaw's reply content. You can run the following program in terminal. AI voice module dialogue does not need to run repeatedly

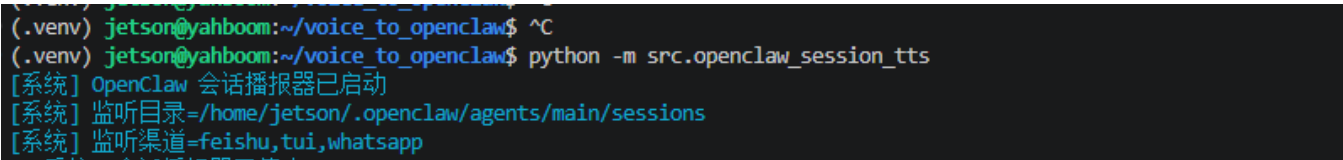
- Modify whether to enable broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter controls whether to broadcast. After modification and saving, run the following program in terminal
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```



5.1 whatsapp Solution

For whatsapp connection steps, please directly refer to [05-Three Dialogue Solution Configuration Tests]. This section only retains dialogue examples suitable for the OLED tutorial.

After creation, you can directly dialogue, for example:

- Display "Welcome to OpenClaw" on OLED
- Help me display current temperature and humidity data on OLED
- Clear OLED screen

19:52



< 2



+86 153 3885 7526 (你)

给自己发消息

with a nice blue glow.

19:44 ✓✓

Display "welcome to openclaw" on
OLED

19:46 ✓✓

Done! "welcome to openclaw" is now
displayed on the OLED screen. The
text was split across two lines due to
the screen's 128-character width
limitation.

19:47 ✓✓

Display "welcome to openclaw" on
OLED

19:52 ✓✓

Done! "welcome to openclaw" is now
displayed on the OLED screen. The
text was split across two lines due to
the 128-character width limit of the
display.

19:52 ✓✓



I

The

Make

Q

W

E

R

T

Y

U

I

O

P

A

S

D

F

G

H

J

K

L



Z

X

C

V

B

N

M



123



space

return



5.2 TUI Solution

If you have completed the TUI self-check in Chapter 0, directly enter the following command in the terminal to continue the dialogue:

```
openclaw tui
```

5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration]. After waking up, you can directly say control content, for example:

- Display hello on OLED
- Change screen content to "Welcome to OpenClaw"
- Clear the screen

```
2026-05-14 17:25:01 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw starting
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 8520 ms 语音, 正在转写

[你] Display high on old.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! "high" is now displayed on the OLED screen. 📺
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Clear the screen.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! The OLED screen has been cleared. 📺
```

6. Comprehensive Cases

Note: The following are all dialogue demonstrations using `openclaw tui` method. You can also choose whatsapp or voice method to complete.

6.1 Case One: Text Display

Experimental Objective

Make OLED display a fixed text to complete the most basic screen control.

Effect Description

After receiving commands, OpenClaw will render the specified text on the OLED screen. This case is suitable for verifying whether text rendering, line breaks, and screen refresh are normal.

Example Instructions

- Display "Hello, OpenClaw" on OLED
- Display "Today's task is testing OLED"
- Change screen content to "System Ready"

6.2 Case Two: Indicator Display

Experimental Objective

Make OLED display single-line status indicators in a more compact way.

Effect Description

This mode is suitable for compressing short information like temperature, humidity, battery level, status values into one line for display, more like a simple dashboard reading.

Example Instructions

- Display "Temp 26C Hum 57%" on OLED
- Write current status as a single line of short information
- Change screen content to "CPU OK RAM OK"

6.3 Case Three: Weather Card

Experimental Objective

Make OLED display a more complete weather information card.

Effect Description

The screen can display information like location, weather, temperature, and humidity. This case is suitable for making weather stations, environmental monitoring, or information panels.

Example Instructions

- Display "Shenzhen Sunny 26°C Humidity 57%" on OLED
- Make a concise weather card
- Format current temperature and humidity into card style display

6.4 Case Four: Animation Display

Experimental Objective

Make OLED play images or animations to display richer visual effects.

Effect Description

Besides static text, OLED can also display bitmaps, GIF animations, or simple self-check animations. This case is very suitable for making boot pages, expression animations, or classroom demonstrations (you can prepare photos or GIF animations to display in the path `/home/jetson/hardware_hal_demo/assets/oled`).

Example Instructions

- Play a simple animation on OLED
- Display an icon image
- Make a boot welcome animation